[INCH-POUND]
A-A-59233
18 May 1998
SUPERSEDING
MIL-E-80270A
25 April 1990

COMMERCIAL ITEM DESCRIPTION

ELECTRICAL DISCHARGE MACHINE, COMPUTER NUMERICAL CONTROL (CNC), WIRE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers computer numerical control (CNC) wire electrical discharge machines (EDMs) and specifies equipment and accessories.

2. CLASSIFICATION. The wire EDMs shall be of the following types and sizes. The type and size of machine to be furnished shall be as specified (see 7.2(b)).

Type I - Submerged
Type II - Nonsubmerged

Size - Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3410

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

TABLE I. Wire EDM sizes.

Size	Type	Minimum machine travel, inches (mm)		
		X-axis	Y-axis	Z-axis
1	I	8.5 (215)	5.5 (140)	4.5 (114)
2		10.0 (254)	6.0 (152)	4.0 (101)
3		12.5 (317)	8.0 (203)	6.5 (165)
4		14.5 (368)	10.5 (267)	8.5 (216)
5		19.5 (495)	12.5 (318)	10.0 (254)
6		23.5 (597)	15.5 (394)	10.5 (267)
7	II	12.0 (304)	9.5 (241)	6.0 (152)
8		13.5 (343)	9.5 (241)	8.5 (216)
9		14.0 (356)	9.5 (241)	7.5 (191)
10		19.5 (495)	13.5 (342)	10.5 (267)

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The wire EDM shall include, as a minimum, the following components:

3.2.1 Worktable. Unless otherwise specified (see 7.2(c)), the worktable length, width, and weight-holding capacity shall be that considered as standard by the manufacturer.

3.2.2 Dielectric system. The dielectric system shall provide recirculating deionized water to the cutting area while maintaining stable operating temperatures. The dielectric shall be cooled by a thermostatically controlled chilling system that regulates the temperature of the dielectric at the workpiece to within 2 degrees of ambient temperature while the machine is cutting. The system shall allow selection of manual or automatic control over the flow of dielectric. The system shall include cartridge filters that filter particles to a 5 micron level. For type I machines, the minimum internal dimensions of the dielectric worktank shall be as specified (see 7.2(d)).

3.2.3 Power supply control unit. The average current capacity of the power supply shall be not less than specified (see 7.2(e)). Unless otherwise specified (see 7.2 (f)), the power supply shall provide for the prevention of dielectric electrolysis.

3.2.4 Automatic wire threading system. Unless otherwise specified (see 7.2(g)), the machine shall be capable of automatically threading wire. In the event of wire breakage, the machine shall automatically rethread the wire and then restart. The machine shall be capable of threading 0.01-inch (0.254-millimeter) wire through a 0.04-inch (1.016-millimeter) hole in a 2-inch (50.8-millimeter) thick workpiece. The machine shall be capable of performing the automatic wire thread function in no more than 1.5 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

3.2.5 Wire guides. When specified (see 7.2(h)), upper and lower wire guides shall be provided to accommodate 0.004-inch (0.102-millimeter), 0.006-inch (0.152-millimeter), 0.008-inch (0.203-millimeter), 0.010-inch (0.254-millimeter), and 0.012-inch (0.305-millimeter) wire, and other sizes as specified.

3.2.6 Spool size. Unless otherwise specified (see 7.2(i)), the wire EDM shall accommodate a minimum wire spool weight of 8 pounds (3.64 kilograms).

3.2.7 Wire chopper. When specified (see 7.2(j)), the wire EDM shall be equipped with a device to chop used wire into small segments to facilitate disposal.

3.2.8 Slug removal. When specified (see 7.2(k)), the machine shall be equipped with a slug removal system.

3.2.9 Battery backup. If required, the CNC system shall have a volatile memory battery backup or uninterrupted power supply (UPS) as specified (see 7.2(l)).

3.2.10 CNC system. The CNC system shall be capable of continuous path, 4-axis (X, Y, U, and V), linear and circular interpolation, with provisions for independent U-V programming to allow machining of workpieces with dissimilar shapes on top and bottom surfaces. Unless otherwise specified (see 7.2(m)), the maximum taper angle requirements per specified workpiece thickness shall be that considered as standard by the manufacturer. The CNC system shall provide adaptive control of machining operations by monitoring the spark gap parameters and pulse width in response to machining conditions. A collision protection system shall be incorporated in the system. The controlled axes shall be identified in accordance with Electronic Industries Association (EIA) 267. When specified (see 7.2(n)), the system shall provide simultaneous control of Z-axis motion.

3.2.10.1 Hardware features. Unless otherwise specified (see 7.2(o)), the CNC system shall include, as a minimum, the following hardware features:

a. Video display terminal.
b. 250-MB hard disk drive.
c. 3.5-inch floppy disk drive.
d. EIA RS-232 interface.
e. Keypad for manual data input.

3.2.10.2 Control panel. Unless otherwise specified (see 7.2(p)), the control panel shall include, as a minimum, the following features:

a. Emergency stop button.
b. Power on/off switches.
c. Jog and jog direction.
d. Feed hold.

3.2.10.3 Operating modes. Unless otherwise specified (see 7.2(q)), the CNC system shall include, as a minimum, the following operating modes:

a. Manual data input.
b. Tutorial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

c. Conversational/interactive mode - The conversational mode shall provide the operator the ability to load an existing program, create a new program, and edit an existing program by following on-screen prompts. As a minimum, the conversational mode shall include prompts for axis and tool management data and part geometry data for creating a part program.
d. Machine diagnostics and monitoring.
e. Dry run.
f. Part program editing.

3.2.10.4 Axis management. Unless otherwise specified (see 7.2(r)), the CNC system shall include, as a minimum, the following axis management features:

a. Backlash compensation.
b. Pitch error compensation.
c. Offset machining.
d. Automatic vertical wire alignment.
e. Automatic hole center and edge positioning.

3.210.5 Software features. Unless otherwise specified (see 7.2(s)), the CNC system shall include, as a minimum, the following software features:

a. Linear and circular interpolation.
b. Inch/metric switchable programming.
c. Floating decimal point.
d. 3-D graphic display.
e. Automatic zero point.
f. Standard G codes, M codes, and canned cycles.
g. Incremental and absolute coordinate positioning.
h. Scaling function for X-, Y-, and Z-axes.
i. Axis rotation.
j. Axis transposing.

3.2.11 Electrical system. Unless otherwise specified (see 7.2(t)), the electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79. The machine shall operate from a 230/460-volt, 3-phase, 60-hertz circuit and shall be wired for the voltage specified (see 7.2(u)). The CNC system shall operate on a 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.12 Electric motors. All wire EDM drive motors shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-7. Unless otherwise specified (see 7.2(v)), the motors shall have open dripproof enclosures.

3.2.13 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(w)).

3.3 Performance. The wire EDM shall meet the performance requirements listed in table I when tested in accordance with sections 5.3.1–5.3.3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

TABLE II. Performance requirements.

Parameter	Tolerance, inches (millimeters)	ISO 230-1 paragraph	ISO 230-4 paragraph
Flatness of work-holding surface	0.0010 (0.03)	5.322 - 5.324	NA
Straightness of X-axis movement	0.0006 (0.015)	5.232.1, 5.232.11	NA
Straightness of Y-axis movement	0.0006 (0.015)	5.232.1, 5.232.11	NA
Parallelism between work-holding frame and X-axis movement	0.0006 (0.015)	5.422.21, 5.422.22	NA
Parallelism between work-holding frame and Y-axis movement	0.0006 (0.015)	5.422.21, 5.422.22	NA
Parallelism between locate pins or work-holding surface and X- axis movement	0.0006/0.0016 (0.015/0.04)	5.422.21, 5.422.22	NA
Parallelism between locate pins or work-holding surface and Y-axis movement	0.0006/0.0016 (0.015/0.04)	5.422.21, 5.422.22	NA
Squareness between X-axis and Y-axis movement	0.0006 (0.015)	5.512.3	NA
Squareness between Z-axis and X-axis movements	0.0008 (0.02)	5.522.2	NA
Squareness between Z-axis and Y-axis movements	0.0008 (0.02)	5.522.2	NA
Parallelism between X-axis and U-axis movement in ZX-plane	0.0010 (0.03)	5.232.1, 5.232.11	NA
Parallelism between X-axis and U-axis movement in XY-plane	0.0006 (0.015)	5.232.1, 5.232.11	NA
Parallelism between Y-axis and V-axis movement in YZ-plane	0.0010 (0.03)	5.232.1, 5.232.11	NA
Parallelism between Y-axis and V-axis movement in XY-plane	0.0006 (0.015)	5.232.1, 5.232.11	NA
Bidirectional accuracy of X-, Y-, and Z-axis movements by numerical control for measured lengths of 19.7, 39.4, and 78.7 inches (500, 1000, and 2000 millimeters)	0.0006/0.0008/0.001 (0.016/0.02/0.025)	2.322.1	NA
Unidirectional accuracy of X-, Y-, and Z-axis movements by numerical control for measured lengths of 19.7, 39.4, and 78.7 inches (500, 1000, and 2000 millimeters)	0.0003/0.0004/ 0.0005 (0.008/0.01/0.013)	2.322.1	NA
Bidirectional accuracy of U- and V-axis movements by numerical control for measured lengths of 3.94 and 7.87 inches (100 and 200 millimeters)	0.0008/0.001 (0.02/0.025)	2.322.1	NA
Undirectional accuracy of U- and V-axis movements by numerical control for measured lengths of 3.94 and 7.87 inches (100 and 200 millimeters)	0.0004/0.0005 (0.01/0.013)	2.322.1	NA
Roundness and axial squareness of machined hole	0.0008/0.0004 (0.02/0.01)	6.622	3.5
Circular hysteresis and circular deviation	0.0008/0.0006 (0.02/0.015)	6.631 - 6.633	3.3, 4.4, 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

3.4 Machine dimensions and weight. If required, the maximum machine dimension (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(x)).

3.5 Safety and health requirements. The manufacturer shall ensure the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(y)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(z)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(aa)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
e. Machine model designation.
f. Power input (volts, total amps, phase, frequency).
g. Short circuit/over-current rating.
h. Contract number or order number.
i. National stock number or plant equipment code.
j. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(bb)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The wire EDM shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(cc)), the wire EDM shall be tested in accordance with sections 5.3.1–5.3.3.3.

5.3.1 Test conditions. The temperature of the environment in which the machine is tested shall be 65° to 78° Fahrenheit (18.3° to 25.6° Celsius). The machine shall be permitted to stabilize for 24 hours prior to the test. The temperature of the environment in which the test is conducted shall not be permitted to vary by more than 2.0° Fahrenheit (1.1° Celsius) during the stabilization period and throughout the test.

5.3.2 Operational test. The wire EDM shall be operated in accordance with the manufacturer's standard operating procedures. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. Unless otherwise specified (see 7.2(dd)), the wire EDM shall be tested in accordance with paragraphs 5.3.3.1 - 5.3.3.3 herein.

5.3.3.1 Roundness and axial squareness of machined hole. The roundness and axial squareness of a machined hole shall be measured in accordance with paragraph 6.622 of ISO 230-1. A 1.3780-inch (35-millimeter) diameter hole shall be cut from a square piece of D2 steel that is 3.1496 inches (80 millimeters) long, 3.1496 inches (80 millimeters) wide, and 1.5748 inches (40 millimeters) thick. The hole shall be cut using 0.008-inch (0.2-millimeter) or 0.012-inch (0.3-millimeter) diameter wire. The average roughness of the finished hole shall be 2 micrometers or better when tested in accordance with ANSI B46.1. The radial deviation of the hole shall not exceed 0.0008 inch (0.02 millimeter) when measured in accordance with paragraph 3.5 of ISO 230-4. The center of the least-squares circle shall be determined at locations 0.197 inch (5 millimeters) and 1.338 inches (35 millimeters) from the top surface of the workpiece. The XY positions of the centers of the least-squares circles shall be no farther apart than 0.0004 inch (0.01 millimeter).

5.3.3.2 Circular hysteresis and circular deviation. Circular hysteresis and circular deviation shall be checked in accordance with paragraphs 6.631 through 6.633 of ISO 230-1 and paragraphs 3.3, 3.4, 4.4, and 6 of ISO 230-4. Depending on the size of the machine, one of the following feed rates and diameters shall be used to perform the tests: A 1.97-inch-per-minute (50-millimeter-per-minute) feed rate shall be used with a 1.57-inch (40-millimeter) diameter. A 2.75-inch-per-minute (70-millimeter-per-minute) feed rate shall be used with a 3.14-inch (80-millimeter) diameter. A 3.94-inch-per-minute (100-millimeter-per-minute) feed rate shall be used with a 6.30-inch (160-millimeter) diameter. A 5.51-inch-per-minute (140-millimeter-per-minute) feed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

rate shall be used with a 12.6-inch (320-millimeter) diameter. The circular hysteresis shall be not greater than 0.0008-inch (0.02 millimeter) and the circular deviation shall be not greater than 0.0006-inch (0.015 millimeter).

5.3.3.3 Cutting speed. The machine shall cut 50.8-millimeter thick D2 steel at a rate of not less than 17 square inches per hour (10,968 square millimeters per hour) using 0.01-inch (0.254-millimeter) uncoated brass wire.

5.4 Acceptance. Unless otherwise specified (see 7.2(ee)), preliminary and final acceptance testing shall be conducted at the manufacturer's site. Failure of the machine to meet any of the performance requirements in sections 5.3.3 through 5.3.3.3 shall be cause for rejection.

6. PACKAGING. Preservation, packing, and marking shall be as specified (see 7.2(ff)).

7 NOTES

7.1 Source of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B46.1	Surface Texture

Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10018.

Electronic Industries Association (EIA)

RS-232	Interface Between Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange
RS-267	Axis and Motion Nomenclature for Numerically Controlled Machines
RS-274	Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines

Application for copies should be addressed to the Electronic Industries Association, 2500 Wilson Boulevard, Arlington, VA 22201-3834.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

International Organization for Standardization (ISO)

ISO 230-1	Test code for machine tools - Part I: Geometric accuracy of machines operating under no-load or finishing conditions
ISO/FDIS 230-2	Test code for machine tools - Part 2: Determination of accuracy and repeatability of positioning of numerically controlled axes
ISO/FDIS 230-4	Test code for machine tools - Part 4: Circular tests for numerically controlled machine tools

Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10018.

National Electrical Manufacturer's Association (NEMA)

NEMA MG-7	Motion/Position Control Motors and Controls

Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 North 17th Street NW, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA-79	Electrical Standard for Industrial Machinery

Applications should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

Occupational Safety and Health Administration (OSHA)

OSHA 1910	General Industry, Safety, and Health Standards (29 CFR 1910)

Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

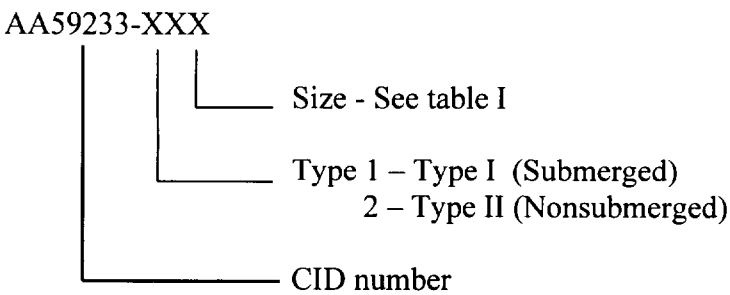
7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type and size required (see 2).
c. Worktable, if different (see 3.2.1).
d. Worktank internal dimensions, as specified (see 3.2.2).
e. Power supply average current capacity, as specified (see 3.2.3).
f. Power supply antielectrolysis capability, if different (see 3.2.3).
g. Automatic wire threading, if different (see 3.2.4).
h. Wire guides, if required (see 3.2.5).
i. Spool size, if different (see 3.2.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

j. Wire chopper, if required (see 3.2.7).
k. Slug removal system, if required (see 3.2.8).
l. Battery backup or UPS, as specified (see 3.2.9).
m. Taper angle per workpiece thickness, if different (see 3.2.10).
n. Simultaneous Z-axis control, if required (see 3.2.10).
o. Hardware features, if different (see 3.2.10.1).
p. Control panel, if different (see 3.2.10.2).
q. Operating modes, if different (see 3.2.10.3).
r. Axis management features, if different (see 3.2.10.4).
s. Software features, if different (see 3.2.10.5).
t. Electrical system requirements, if different (see 3.2.10).
u. Operating voltage, as specified (see 3.2.11).
v. Motor enclosure, if different (see 3.2.12).
w. Optional accessories, as specified (see 3.2.13).
x. Machine dimensions and weight, as specified (see 3.4).
y. System of units required (see 3.6).
z. Dual system of units, if required (see 3.6).
aa. Nameplate, if different (see 3.7).
bb. Lubrication plate or chart, if required (see 3.8).
cc. Testing, if different (see 3.5).
dd. Performance test requirements, if different (see 5.3.3).
ee. Acceptance, if different (see 5.4).
ff. Packaging requirements, as specified (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians
Army - AL
Air Force - 99
Navy - SH

Reviewers
Air Force - 84
Army - AR, AV, CR4
Navy - MC, SA

PREPARING ACTIVITY:
DLA - GS

(Project 3410-0060)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:06Z
Check the source to verify that this is the current version before use.